



A game theoretic approach for time-of-use pricing with considering renewable portfolio standard effects and investment in energy storage technologies under government interventions

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ABSTRACT

To remain in the competitive market, power companies, in addition to price competition, need to develop technologies to improve the reliability of electricity supply and achieve sustainable development goals. This paper explores energy pricing in an electricity supply chain consisting of renewable and conventional electricity producers and a distribution grid, focusing on improving supply reliability and achieving sustainable development goals. The study employs game theory to model government intervention in the electricity market through subsidization and Renewable Portfolio Standard (RPS) strategies. Three investment scenarios for improving reliability are considered, with producers determining wholesale prices and the distribution grid setting prices for low-and-high-load periods. The renewable producer and the distribution grid manage the reliability level by investing in energy storage technologies in the first and second scenarios, respectively, while it is ignored in the third scenario. The results reveal that optimal subsidy and RPS rates depend on the government's priorities regarding environmental effects, government revenue, and social welfare. Scenario two, is found to be the most favorable for improving supply reliability and achieving social welfare and environmental goals. Additionally, prioritizing social welfare maximization policy is found to be advantageous for producers' profit, the distribution grid, and supply reliability improvement.

1. Introduction

Integrating renewable and conventional energy sources can deliver several benefits, such as higher cost-effectiveness, and reduced environmental effects [1]. International Renewable Energy Agency lays out a roadmap, which recommends that the share of renewable sources in total energy production is expected to increase from about 25% in 2017 to about 86% in 2050 [2].

A typical electricity system has three main stakeholders: the government, producers, and consumers [3,4]. Governments have a critical role in achieving sustainable development objectives in electricity supply chains with different mechanisms [5,6]. Governments often use subsidies and renewable portfolio standard (RPS) mechanisms to pursue sustainable development objectives, including increasing social welfare and reducing adverse environmental impacts of sustainable development [7,8]. In other words, increasing the share of renewable sources in a country's total energy composition can contribute to its sustainable

development [9]. RPS is a quantity-based support scheme whereby electricity suppliers are obligated to get a fixed portion of all electricity they sell from renewable sources [10,11]. RPS can significantly affect electricity prices and profits of members of an electricity supply chain [12]. Governments can also use subsidies to develop renewable technologies, promote competitiveness in the renewable industries, increase the acceptance of renewable-assisted technologies among consumers, and accelerate the integration of renewable sources with other energy systems [13]. The ultimate goals are reducing average production costs through renewable electricity generation or achieving certain environmental and social objectives [14].

Meanwhile, with the rising demand for electricity and the increasing dependence of humans across the world on stable electricity access, continuous efforts are needed to ensure an appropriate level of reliability in the power supply [3]. According to Yaqoot et al. [9], one of the significant technical obstacles to developing renewable sources is the low reliability of these sources because of their intermittent nature. The solutions proposed for overcoming this obstacle and improving

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Abbreviations			
RPS	Renewable Portfolio Standard	L_R	The lower bound revenue for the government (\$)
TOU	Time-Of-Use	L_{EC}	The upper bound of environmental impact for the government (TonCO ₂)
FIT	Feed-in tariff	L_S	The lower bound of social welfare for the government (\$)
Indices		Decision Variables	
i	Index of low load and high load periods $i \in \{l, h\}$	λ_k	Renewable portfolio standard proportion under scenario k
k	Scenario $k \in \{1, 2, 3\}$	s_k	The considered subsidy by the government under scenario k
l	Low load period	w_k	Wholesale price of conventional energy under scenario (\$/Mwh)
h	High load period	τ_k	Wholesale price of renewable energy under scenario k (\$/Mwh)
j	Producers, Distribution Grid ($j \in \{r, n, GRID\}$)	p_{ik}	Selling price of distribution grid in period. i under k scenario (\$/Mwh)
Parameters		R_k	Reliability in the k scenario
α_i	Market base for the energy in period i (Mwh)	Demand and profit functions	
ρ	Percentage of subsidy that government give to the renewable energy producer	D_{ik}	Demand in period. i under scenario k (Mwh)
β	Self-price sensitivity coefficient (Mwh/\$)	π_{Gridk}	Profit function of distribution grid under scenario k (\$)
φ	The economic value of generating each unit of renewable electricity for the government	π_{rk}	Profit function of renewable energy producer under scenario k (\$)
Q_j	Environmental impacts of producer j (TonCO ₂ /Mwh)	π_{nk}	Profit function of conventional energy producer under scenario k (\$)
θ_2	Elasticity of the demand function for the high-load period relative to the specified price for the other period (Mwh/\$)	π_{swk}	Government function for social welfare goal under scenario k (\$)
θ_1	The elasticity of the demand function for the low-load period relative to the specified price for the other period (Mwh/\$)	π_{GOVk}	Government function for revenue generation goal under scenario k (\$)
c_j	Cost of generating one unit of electricity by the producer j (\$/Mwh)	π_{ERk}	Government function for adverse environmental impacts goal under scenario k (TonCO ₂)
μ	Reliability sensitivity coefficient		
f	The percentage of renewable energy producer's cooperation in the storage technologies expenditure		

reliability include the development of better power storage technologies and adoption of more effective demand-side management strategies [3, 15]. One of these solutions involves investing in energy storage technologies and equipment to minimize fluctuations in electricity flow from renewable sources so that the electricity supply-demand balance can be maintained consistently, leading to the improved reliability and sustainability of renewable power systems [3, 16]. The other solution is to utilize demand-side management strategies, including the Time-Of-Use (TOU) pricing scheme. It is an effective way to reduce operating costs (by up to 6.06%) as well as load optimization (decreasing peak demand or shifting the load to off-peak periods) [17, 18].

To the best of the authors' knowledge, no previous study has clarified the subject of electricity pricing for demand-side management while considering the effects of electricity supply reliability and government interventions to promote the production and use of renewable energies in a competitive electricity supply chain. In other words, this is the first study to explore the impact of the RPS ratio and subsidy on demand and renewable/conventional electricity wholesale prices under a competitive environment in the energy market. Another novel aspect of this study is the inclusion of electricity supply reliability by investing in energy storage technologies during TOU pricing. Also, we use the game-theoretic approach to examine the mentioned factors under government interventions to achieve sustainable development objectives in electricity supply chains.

This paper implements a game-theoretic approach to explore the influences of electricity supply reliability improvement through investment in energy storage technologies, TOU pricing, and RPS on the decisions of the members of an electricity supply chain. We suppose that the government intervenes in the market because of the importance of its decisions in promoting development, using renewable energy sources, and pursuing sustainable development objectives. It is assumed that

the government may pursue three goals in these decisions, including minimizing environmental impacts and maximizing government revenue and social welfare. The government is assumed to have two mechanisms for achieving these goals: subsidizing the distribution grid, renewable electricity producers, and enforcing the RPS. Also, the three following scenarios for investment in storage technologies are considered to improve electricity reliability. (i) The acquisition is made jointly by renewable producers and conventional producers, (ii) the investor is the distribution grid, and (iii) no investment is made in storage technologies. Supply chain members make equilibrium decisions according to the government's policy in each scenario.

This research aims to respond to the following questions.

- 1 According to game theory, what would be the optimal pricing scheme under government intervention for supply reliability improvement in an electricity supply chain comprised of conventional and renewable electricity producers and the distribution grid?
- 2 According to game theory, how can utilizing demand-side management in the distribution grid improve the demand load pattern of an electricity supply chain?
- 3 How can the government use subsidy and RPS mechanisms to achieve sustainable development goals? What would be the effect of these mechanisms in each defined scenario on demand, prices, and profit functions? Moreover, which government policy would highly benefit the supply chain members?
- 4 What would be the best scenario for investing in energy storage technologies for reliability improvement? In addition, should the investment be made jointly by conventional and renewable producers or the distribution grid?
- 5 For each scenario and each government policy, what would be the energy demand for low- and high-load periods, selling prices of the

distribution grid in the two periods, wholesale prices and profit of conventional and renewable producers, and the profit of the distribution grid and government functions?

This research responds to the mentioned questions in a hierarchical and dynamic Stackelberg game model. In this game, as the leader at the first level, the government decides the value of subsidies to be paid to the distribution grid and the renewable producer, and the RPS ratio is based on three policies. At the second level, renewable and conventional producers simultaneously determine the wholesale electricity prices (the wholesale price of the electricity sold to the distribution grid) in a competitive environment. At the third level, the distribution grid determines the sell prices of electricity in low- and high-load periods with the objective of demand-side management. Decisions regarding the investment in energy storage technologies for reliability improvement are made in three scenarios: investment by the renewable producer at the second level, investment by the distribution grid at the third level, or neither. Finally, the players' equilibrium decisions after adopting each government policy in each of the three scenarios are compared.

This study investigates the influence of government policies and strategies on the decisions of the members of the electricity supply chain. Also, suggestions based on the results are presented with the aim of achieving sustainable development goals in the field of production and promotion of renewable energy consumption, which can help the government and companies to make policies in the field of renewable energy, and this is not related to a specific company, country, or region. In fact, the results of this research can be used for all countries trying to achieve sustainable development goals.

The rest of this paper is organized as follows: A review of the literature on the subject is provided in Section 2. Section 3 provides a definition of the problem and discusses the modeling approach and assumptions, and equilibrium solutions. A numerical example and its equilibrium solutions followed are provided in section 4. Section 5 explains the obtained results and managerial insights. Finally, the conclusions and future research are presented in Section 6.

2. Literature review

The subject discussed in this research is the pricing in sustainable energy supply chains with government intervention for reliability improvement and demand-side management using a game-theoretic approach. Thus, the relevant literature is reviewed in three parts: reliability and demand-side management in electricity systems, government interventions in developing renewable energy sources, and application of game theory related to energy problems.

2.1. Reliability and demand-side management in electricity systems

Yaqoot et al. [9] reviewed the literature on barriers to developing renewable energy sources and the measures needed to address them. This study stated that one of the measures needed to remove barriers to developing renewable energy sources is investing in energy storage technologies for reliability improvement. Tran et al. [19] introduced an effective solution for integrating renewable energy sources and storage devices into a conventional grid to improve the reliability and efficiency of the electricity system and reduce the total cost of electricity for consumers using TOU pricing. Their findings showed that the energy stored in storage devices and the energy received from the grid could be managed by reducing the electricity consumption in the high price periods.

A study by Zhao et al. [20] analyzed local energy market competition between renewable energy suppliers with and without energy storage. They used a three-stage game theory model to examine equilibrium decisions on investment in storage options in the competition among renewable energy suppliers. Yang et al. [21] presented a model using the game theory for optimizing TOU pricing strategies and showed that it

decreased the electricity company's costs, increased consumers' profits, and improved social welfare. Azad and Ghotbi [22] introduced a Cournot game theory model for an electricity market with suppliers, utility grids, and consumers. In their model, suppliers were assumed to include renewable energy sources, diesel generators, and storage devices. In a study by Adefarati and Bansal [23], a microgrid system's reliability and economic and environmental benefits with renewable energy sources were analyzed. Their results showed that using green technologies in a microgrid system enhances reliability, cost-saving, and environmental effects. Tang et al. [24] used a Stackelberg game model to examine a power network's and buildings' interactive strategies for demand-side management purposes. In this model, price optimization by the network increased profits and reduced demand fluctuations.

2.2. Government interventions in the development of renewable energy sources

Governments can adopt various policies to develop renewable energy sources. Sun and Nie [25] compared two government regulatory policies which helped the renewable energy industry using RPS and feed-in tariff (FIT). Their results showed that RPS is more efficient than the FIT policy in reducing carbon emissions and improving consumer surplus. Myojo and Ohashi [26] analyzed the effects of consumer subsidies in support of renewable sources on industry growth and social welfare. Their results showed that subsidies encourage the use of renewable energy sources.

The effect of government regulations on the production and transmission of renewable power in the Chinese electricity market was studied by Fang et al. [7]. It was found that provincial governments managed to increase the transmission of renewable energy from the grid by providing subsidies. Benton et al. [27] developed a general equilibrium model for the RPS, ultimately reporting that the RPS assists in reducing greenhouse gas emissions. Yi et al. [28] used the evolutionary game theory and the system dynamics approach to analyze the behavior and strategy of conventional and renewable electricity producers under RPS and subsidy schemes. They reported that raising fines increases the willingness of electricity producers to commit to RPS, and RPS should replace FIT in the future.

In a study by Li et al. [29] on the impact of RPS on various objectives of the energy policies of the Chinese government, it was found that the implementation of RPS regulates energy structure and reduces carbon emissions. Xian-gang et al. [30] investigated the effect of RPS and carbon tax policies on the electricity market. Adopting these regulations increased renewables' share in total energy production, thereby reducing carbon emissions due to the electricity market. Zhu et al. [31] presented an equilibrium model for an electricity market regulated by carbon tax policy and RPS. Their results showed that implementing the RPS compels electricity companies to sell renewable energy, thus leading to the development of the renewable electricity industry. Safarzadeh et al. [32] analyzed the behavior of the suppliers in an energy-sustainable supply chain under three Scenarios, including conventional power generation, energy efficiency program, and RPSs. They aim to decide whether to use one of the tradable white certificate markets and the tradable green certificate market under government intervention. Norouzi et al. [33] designed a model that considers the government, energy intermediaries, and the final product firms as the main three actors of the game. The government sets tax rates to maximize social welfare, and firms maximize profits by obtaining tax rates by choosing factors of production. Wang and Li [34] analyzed a bi-level decision-making model for the government and power plants by considering RPS and green certificate trading policy. Their research simulated the pricing of power plants in different scenarios, participation in the green certificate trading market, and decision-making in the carbon trading market. Under government intervention, xin-gang and Yu-Qiao [11] investigated competition between renewable energy and conventional suppliers. Their results showed that with the

implementation of RPS, the uncertainties of power producers' behaviors decreased gradually.

2.3. Application game theory in related to energy problems

Similar to all other markets, almost every electricity market has been competitive because of the increasing number of electricity suppliers in markets. On the other hand, game theory is one of the most effective methods to model any competitive market. In recent years, game theory has been used in many cases, including investigating government policies, pricing in competitive markets, and obtaining equilibrium values for other variables in the electricity supply chain.

Using the Stackelberg game theory, Kok et al. [35] investigated the impact of electricity pricing policies on distributed renewable energy investments under net metering. Liu et al. [14] developed a game theory-based modeling framework for optimizing the design of multi-energy systems and renewable subsidy strategies to meet demand as well as government decarbonization goals. Their results showed that government subsidy policies effectively enhance the penetration of renewable energy. Yao et al. [12] developed a model using the Stackelberg game theory to examine the effect of RPS on company profits and electricity prices in an electricity market, including a producer (producing both renewable and conventional energy), a retail company, and a distribution grid. Jamali and Rasti-Barzoki [36] investigated government support strategies and licensing contracts in an electricity supply chain, including two renewable electricity suppliers and a grid company with technology spillover using the game-theoretic approach. In another study, Jamali et al. [37] addressed the energy resilience role in the competition between conventional and renewable energy producers by applying a game-theoretic approach. Amiri-Pebdani et al. [38] used the game theory to examine competition between conventional and renewable energy supply chains for energy pricing with government intervention to achieve sustainable development goals.

Table 1 provides a summarized description of previous studies in the research literature and compares them with the current study.

To the best of the authors' knowledge, no previous study has clarified the subject of electricity pricing for demand-side management while considering the effects of electricity supply reliability and government interventions to promote the production and use of renewable energies in a competitive electricity supply chain. In summary, the innovations of this study are as follows.

- 1 TOU pricing for demand-side management simultaneously with decision-making on the level of reliability in an electricity supply chain with conventional and renewable producers.
- 2 Exploring the effect of investment of the distribution grid and electricity producers in energy storage technologies to improve the reliability of the electricity supply
- 3 Exploring the effect of government support strategies by paying subsidies to renewable electricity producers and the distribution grid to implement RPS and encourage the distribution grid to supply

electricity from renewable producers and ultimately achieve sustainable development goals.

- 4 Finding and comparing equilibrium solutions for grids' sell prices in low- and high-load periods, reliability level, wholesale prices of conventional and renewable producers, government subsidies, RPS, demand, and profit functions, and providing managerial insights for electricity supply chain decisions and policymakers.

3. Problem description and formulation

The research investigates the impact of government policies and strategies on the decision-making of electricity supply chain stakeholders. Based on the results, recommendations are presented to support the achievement of sustainable development goals in producing and promoting renewable energy consumption. These proposals can assist governments and companies in shaping their renewable energy policies and strategies. It is important to note that the study is not limited to any particular company, country, or region, and the findings can be applied in all countries that aim to attain sustainable development goals.

In this study, TOU pricing is performed for demand-side management, and investment in energy storage technologies is performed to improve the reliability of the electricity supply. Government interventions can lead to achieving sustainability goals in the electricity supply chain. The players in the electricity supply chain include the government and supply chain members (comprising the renewable and conventional electricity producers and the distribution grid). Electricity generated by the two producers is sold to the distribution grid to meet consumer demand in the presence of demand-side management. Two separate demand functions are considered for low- and high-load periods. Demand-side management is performed by changing the pattern of energy consumption in the low- and high-load periods in the demand functions. The investment in energy storage technologies is incorporated into demand functions for higher reliability to increase demand.

Three scenarios are defined for problem modeling. In order to achieve the goals of sustainable development in each scenario, three policies of maximization of government profit, maximization of social welfare, and minimization of environmental effects, and two strategies of subsidy and RPS in the Triple Bottom Line framework are considered. It is assumed that the government pays a subsidy to increase the share of renewables in the total electricity production up to a target ratio (denoted by λ). A part of this subsidy is paid to the electricity production companies to increase renewable electricity generation, and the rest is given to the distribution grid to encourage the purchase of renewable power. The government's decision variables are the value of the subsidy to be paid to supply chain members and the RPS ratio. In the end, we examine the impact of the subsidy value and the RPS ratio on the profits of chain members, wholesale electricity prices, demand, and sustainability factors in each scenario to assist government policymaking.

In the first scenario (Fig. 1(a)), the renewable electricity producer invests in storage technologies to improve the reliability of the electricity supply and customers' tendency to purchase renewable power. Since this investment increases the demand for conventional power, the

Table 1
Comparison of the research literature on the subject with the current research.

Study	energy pricing	Reliability level	Demand side management	Government intervention		Renewable energy	Conventional energy	Competition	Distribution grid as player	Game Theory
				Subsidy	RPS					
[10]					✓	✓	✓			✓
[12]	✓				✓	✓	✓		✓	✓
[22]	✓	✓				✓				✓
[35]	✓		✓	✓		✓	✓			✓
[36]	✓	✓		✓		✓		✓	✓	✓
[38]	✓		✓	✓		✓	✓	✓		✓
current Research	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

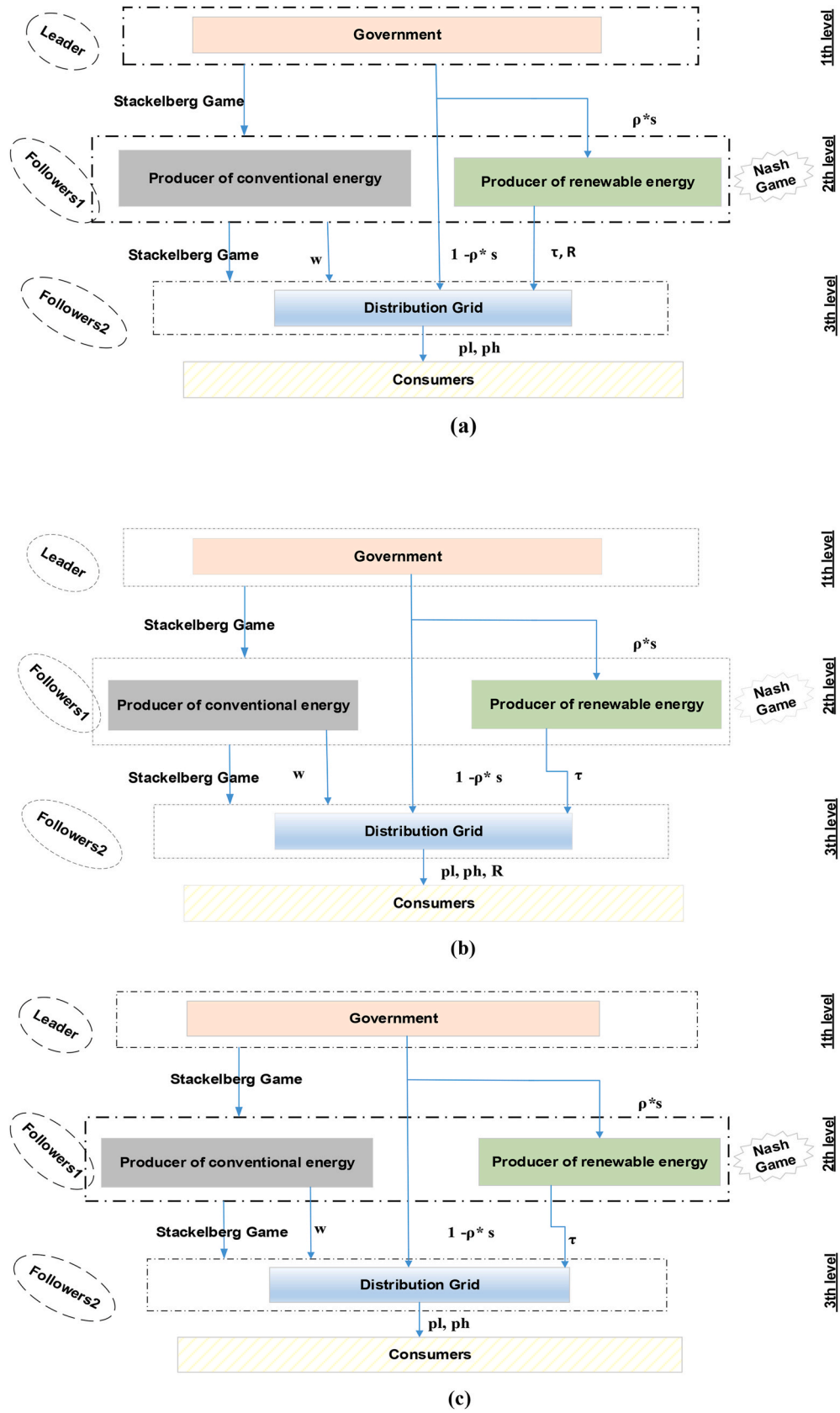


Fig. 1. Problem structure in (a) Scenario 1, (b) Scenario 2, (c) Scenario 3.

conventional electricity producer also participates in the acquisition by paying a percentage of the cost of developing and implementing energy storage technologies. In this scenario, there is assumed competition between two producers regarding wholesale prices. The decision variable for the conventional producer is the wholesale price of electricity. For the renewable producer, the decision variables are the wholesale price of renewable electricity and the investment in energy storage technologies for reliability improvement. The distribution grid must decide the selling price of electricity in low- and high-load periods with the goal of demand-side management. In the second scenario (Fig. 1(b)), the distribution grid invests in energy storage technologies for reliability improvement. Therefore, in this scenario, the deciding variables of the distribution grid are the supply reliability and the selling price of electricity in both low- and high-load periods. In contrast, conventional and renewable producers' decision variables are wholesale electricity prices. In the third scenario (Fig. 1(c)), electricity pricing for low-load and high-load periods is performed without considering the reliability of the electricity supply.

The problem is modeled using game theory to analyze players' behaviors and interactions. The game is assumed to have a leader-follower structure and is solved using backward induction. In every scenario, the government acts as the leader, conventional and renewable electricity producers (with a Nash competition between them) act as the first follower, and the distribution grid acts as the second follower (Fig. 1. (a), (b), (c)).

We follow the systems thinking-based system dynamics approach [39]. Then, we draw cause-and-effect relationships of problem variables in the first scenario, which are illustrated in Fig. 2. It should be noted that a positive causal link in this diagram means that two nodes change in the same direction. In addition, a negative causal link implies that the two nodes change in opposite directions. As shown in Fig. 2, by increasing the investment in storage equipment, the electricity supply reliability and, consequently, the demand in low- and high-load periods, and the profit of the distribution grid and the renewable producer will increase. Increasing the RPS ratio increases the demand for renewable energy and decreases the demand for conventional energy. The increased demand for renewable energy leads to increased profits for the renewable electricity producer and reduced profits for the conventional electricity producer, helping the government achieve the Triple Bottom Line goals. As the value of the subsidy that the government pays to the distribution grid and the renewable energy producer increases, the distribution grid's willingness to purchase renewable energy, the profits of the renewable electricity producer, and the social welfare increase.

Furthermore, the adverse environmental effects and the government revenue decrease.

Meanwhile, changes in demand in both low- and high-load periods reflect the effect of demand-side management. Thus, both government policies, directly and indirectly, affect the economic, social, and environmental indicators and thus achieve sustainable development objectives. This study reveals the optimal solutions for the decision variables previously described. It analyzes equilibrium solutions to reach some insights for the members of electricity supply chains and government policymakers. The problem is modeled with the following assumption.

Assumption 1. The total amount of electricity sold by the distribution grid equals the demand for electricity in low- and high-load periods.

Assumption 2. Demand in low- and high-load periods depends on the supply reliability as well as the grid's selling price in low-load and high-load periods. The price dependence coefficient of each period is higher than the corresponding coefficient of the other period ($\beta > \theta_1, \beta > \theta_2$).

Assumption 3. Reliability level affects demand less than price ($\beta > \mu$).

Assumption 4. The investment cost is considered as a function of the level of reliability ($\psi \frac{(R_k)^2}{2}$) ([36]).

Assumption 5. The unit cost of renewable energy production is higher than the unit cost of conventional energy production ($c_r < c_n$) [12].

Assumption 6. The distribution grid purchases its required electricity from the renewable producer in the ratios of λ_k and the conventional producer in the ratios of $1 - \lambda_k$.

3.1. Modeling

3.1.1. Demand functions

In this problem, the power supply chain comprises a renewable electricity producer, a conventional electricity producer, and the electricity distribution grid. In this chain, the main goal is to meet demand with an appropriate level of reliability with TOU pricing in place for demand-side management through the distribution grid and the government intervening to improve production and use of renewable power for sustainable development purposes. Three scenarios are considered for modeling. In the first scenario, conventional and renewable producers jointly invest in energy storage technologies. The produced energy is sent to the distribution grid for demand-side management. In this problem, demand functions are not only a linear function of price in low-

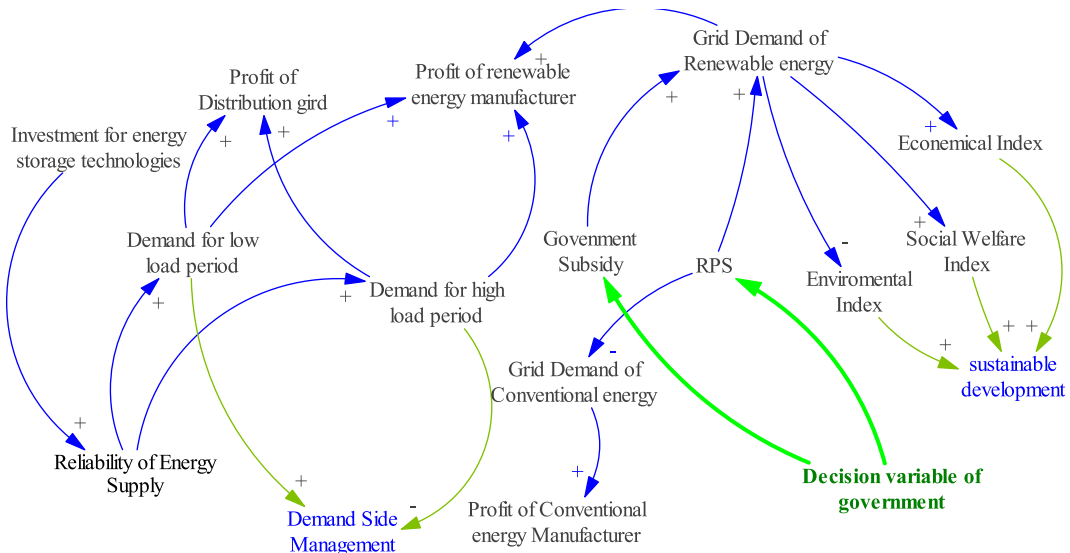


Fig. 2. Cause-and-effect relationship of system dynamics framework for the problem under scenario 1.

and high-load periods [35,38] but also depend on supply reliability. For all three scenarios, demand functions are formulated as Eqs. (1) and (2). Unlike the two first scenarios, in scenario 3, R_3 is considered zero.

$$D_{lk} = \alpha_l - \beta p_{lk} + \theta_1 p_{hk} + \mu R_k \quad (1)$$

$$D_{hk} = \alpha_h - \beta p_{hk} + \theta_2 p_{lk} + \mu R_k \quad (2)$$

Notably, in Eq. (1), which computes demand in the low-load period, the first term (α_l) is the potential demand in the low-load period. The second term (βp_{lk}) is the demand sensitivity for the low-load period. The third term ($\theta_1 p_{hk}$) and the fourth term (μR_k) are the demand sensitivity for the high-load period and the demand sensitivity to the electricity supply reliability Eq. (2) computes the demand in the high-load period.

3.1.2. Goal functions of players for scenario 1

The profit functions of the players in the first scenario are provided as follows. The profit of the distribution grid is computed using Eq. (3). In this equation, λ_k is the target RPS ratio in scenario k [12]. It is assumed that the distribution grid purchases its required electricity from the renewable producer at a price τ_k in the ratios of λ_k and the conventional producer at a price w_k and in the ratios of $1 - \lambda_k$. For demand-side management, the distribution grid sells the power to the consumer at prices p_{hk} and p_{lk} for high- and low-load periods, respectively. The government pays a subsidy $((1 - \rho)s_k)$, to the distribution grid to encourage it to buy renewable power. The profit functions of conventional and renewable electricity producers are given in Eqs. (4) and (5), respectively. In Eq. (4), the first term is the profit that the conventional electricity producer earns by providing $1 - \lambda_k$ of the distribution grid's supply, and the second term is the cost incurred by this producer due to its participation in the investment in energy storage technologies. Since the investment of the renewable electricity producer in storage technologies can ultimately enhance the supply reliability of the conventional electricity producer, hence, the conventional electricity producer bears $1 - f$ portion of the cost of investment. Following the approach of Allan [40], the investment cost function is considered quadratic. In Eq. (5), the first term is the profit that the renewable electricity producer earns by selling power to the distribution grid plus the subsidy receives from the government as a motive to produce more renewable energy (ρs_k), and the second term is the cost it incurs due to investment in energy storage technologies for supply reliability improvement.

$$\pi_{Grid1} = (p_{l1} - (\lambda_1(\tau_1 - (1 - \rho)s_1)) - (1 - \lambda_1)w_1)D_{l1} + (p_{h1} - (\lambda_1(\tau_1 - (1 - \rho)s_1)) - (1 - \lambda_1)w_1)D_{h1} \quad (3)$$

$$\pi_{n1} = (w_1 - c_n)(1 - \lambda_1)(D_{l1} + D_{h1}) - \psi(1 - f)\frac{(R_1)^2}{2} \quad (4)$$

$$\pi_{r1} = (\tau_1 + \rho s_1 - c_r)(\lambda_1)(D_{l1} + D_{h1}) - \psi f \frac{(R_1)^2}{2} \quad (5)$$

3.1.3. Goal functions of players for scenario 2

In the second scenario, it is assumed that the distribution grid invests in energy storage technologies. In this scenario, demand functions are in the form of Eqs. (1) and (2) and the profit functions of the distribution grid, conventional and renewable electricity producers are in the form of Eqs. (6)–(8). In Eq. (6), the last term is the investment cost incurred by the distribution grid, which is considered a quadratic function.

$$\pi_{Grid2} = (p_{l2} - (\lambda_2(\tau_2 - (1 - \rho)s_2)) - (1 - \lambda_2)w_2)D_{l2} + (p_{h2} - (\lambda_2(\tau_2 - (1 - \rho)s_2)) - (1 - \lambda_2)w_2)D_{h2} - \psi \frac{(R_2)^2}{2} \quad (6)$$

$$\pi_{n2} = (w_k - c_n)(1 - \lambda_2)(D_{l2} + D_{h2}) \quad (7)$$

$$\pi_{r2} = (\tau_2 - c_r + (\rho)s_2)(\lambda_2)(D_{h2} + D_{l2}) \quad (8)$$

3.1.4. Goal functions of players for scenario 3

In the third scenario, it is assumed that none of the supply chain members invest in energy storage technologies. In this scenario, demand functions are in the form of Eqs. (9) and (10). Unlike the two previous scenarios, no reliability term is in these equations. Similarly, the investment cost in energy storage technologies is not included in the profit functions given in Eqs. 11–13.

$$D_{l3} = \alpha_l - \beta(p_{l3}) + \theta_1 p_{h3} \quad (9)$$

$$D_{h3} = \alpha_h - \beta(p_{h3}) + \theta_2 p_{l3} \quad (10)$$

$$\pi_{Grid3} = (p_{l3} - (\lambda_3(\tau_3 - (1 - \rho)s_3)) - (1 - \lambda_3)w_3)D_{l3} + (p_{h3} - (\lambda_3(\tau_3 - (1 - \rho)s_3)) - (1 - \lambda_3)w_3)D_{h3} \quad (11)$$

$$\pi_{n3}[w_3] = (w_3 - c_n)(1 - \lambda_3)(D_{l3} + D_{h3}) \quad (12)$$

$$\pi_{r3}[\tau_3] = (\tau_3 + \rho s_3 - c_r)(\lambda_3)(D_{l3} + D_{h3}) \quad (13)$$

3.1.5. Government

Three government goals are formulated based on Eqs. 14–16 to achieve sustainable development. In Eq. (14), it is assumed that the government benefits from φ units of economic value for the production of each unit of renewable electricity. It is due to benefits such as reduced carbon emission, abundant power supply, job creation, etc. However, to support the production and consumption of renewable energy, the government subsidizes the renewable producer and the distribution grid by a total of s_k units per unit of renewable electricity generation.

As the social welfare function is considered to be a function of the consumer surplus [41,42], it is formulated in Eq. (15). Consumer surplus is calculated based on the difference between the market price of electricity in each period and the consumer's willingness to pay [38,42–44]. We follow Hafezalkotob [45] to formulate the function of environmental impacts (Eq. (16)). Where, Q_r and Q_n are the adverse environmental impacts of renewable and conventional energies, respectively.

$$\pi_{GOV}[s_k, \lambda_k] = (\varphi - s_k) * (\lambda_k) * (D_{lk} + D_{hk}) \quad (14)$$

$$\pi_{SWk} = \int_{p_{ij}}^{\frac{\alpha_l + p_{lk} + \theta_1 + \mu R_k}{\beta}} D_{lk} dp_{lk} + \int_{p_{hk}}^{\frac{\alpha_h + p_{hk} + \theta_2 + \mu R_k}{\beta}} D_{hk} dp_{hk} \quad (15)$$

$$\pi_{ERk} = Q_r * (\lambda_k) * (D_{lk} + D_{hk}) + Q_n * (1 - \lambda_k) * (D_{lk} + D_{hk}) \quad (16)$$

The model is solved for each of these objectives. One goal is considered an objective function, and the two other goals are considered constraints to calculate the subsidy amount and the RPS ratio for the three policies given in Table 2. In Policy 1, the government revenue maximizes considering the lower bound of social welfare and the upper bound of environmental effects. Policy 2 maximizes social welfare by considering government revenue's lower bound and adverse environmental impacts' upper bound. Policy 3 minimizes the adverse environmental impacts for the lower social welfare and government revenue bounds.

Table 2
Government policies.

Policy	Goal	Constraints	Scenario k
1	π_{GOV}	π_{SW} π_{ER}	$\left\{ \begin{array}{l} \text{Max } \pi_{RG}(s_k^*, \lambda_k^*) \\ P_{ik}, w_k, \tau_k, R_k \\ \pi_{RGK} \geq L_R \\ \pi_{ERK} \leq L_E \end{array} \right.$
2	π_{SW}	π_{GOV} π_{ER}	$\left\{ \begin{array}{l} \text{Max } \pi_{SW}(s_k^*, \lambda_k^*) \\ P_{ik}, w_k, \tau_k, R_k \\ \pi_{RGK} \geq L_R \\ \pi_{ERK} \leq L_E \end{array} \right.$
3	π_{ER}	π_{GOV} π_{SW}	$\left\{ \begin{array}{l} \text{Min } \pi_{ER}(s_k^*, \lambda_k^*) \\ P_{ik}, w_k, \tau_k, R_k \\ \pi_{RGK} \geq L_R \\ \pi_{SWK} \geq L_S \end{array} \right.$

3.2. Optimal equilibrium solutions of scenarios

In each scenario, we first obtain the equilibrium solutions of the distribution grid (as the second follower). Then, using backward induction, the solutions of the second level are inserted into the profit functions of conventional and renewable producers (as the first follower). After that, the best equilibrium solutions for the producers are obtained using simultaneous solutions. Finally, all obtained solutions are inserted into the functions of the government (as the game leader) to determine the optimal amount of subsidy and RPS. The game model of the problem in the proposed scenarios is presented in Table 3.

3.2.1. Equilibrium solutions of scenario 1

A Stackelberg game is applied to obtain the equilibrium solutions. In each scenario, the game leader is the government, the first followers are the producers, and the second follower is the distribution grid. In the sequence of decisions in scenario 1, the distribution grid, as the second follower, determines the selling prices of electricity in low- and high-load periods in the first stage. In the second stage, by playing a Nash game between the producers, the wholesale electricity price for each producer and the electricity reliability are determined simultaneously. In the third stage, the government sets the equilibrium of its subsidy and RPS under each policy using backward induction. Equilibrium solutions of scenario 1 are given in Theorem 1. The proofs of this theorem and all other theorems are presented in Appendix A and Appendix B.

Theorem 1. In scenario 1, the optimal decisions on the wholesale

$$\tau_2 = \frac{(M5 + M1((-1 + \lambda)c_n + 2\lambda c_r) + M4 + (2H9s\lambda(-1 + 3\rho) + M2\alpha_h - H8\alpha_l)\theta_2 - M4\theta_2^2)}{3M1\lambda} \quad (26)$$

prices of conventional and renewable electricity, the reliability level, and the distribution grid's selling prices in low-load and high-load periods are given in Eqs. (17)–(21).

$$p_{l1} = \left(-\frac{Q0}{Q2} + \frac{Q1}{2\beta + \theta_1 + \theta_2} \right) \quad (17)$$

$$p_{h1} = Q3 + \frac{Q0}{4\beta - 2\theta_1 - 2\theta_2} - \frac{Q1}{2(2\beta + \theta_1 + \theta_2)} \quad (18)$$

$$w_1 = \frac{\frac{fK2\psi}{-1+\lambda} - K1c_n}{K3} \quad (19)$$

$$\tau_1 = -\frac{K6s\lambda + fK4\psi + K1\lambda c_r + 2f(-1 + \lambda)\psi c_n(\beta^2 - \theta_1\theta_2)}{K3\lambda} \quad (20)$$

$$R_1 = \frac{\mu(-K5\beta - \alpha_l\theta_2 + \theta_1(-\alpha_h + 2((-1 + \lambda)c_n + \lambda(s - c_r))\theta_2))}{K3} \quad (21)$$

Where the values of Q_0 – Q_3 and k_1 – k_6 are given in Appendix C.

Table 3
Game model for the problem.

Scenario 1	Scenario 2	Scenario 3
$\left\{ \begin{array}{l} \text{Max(Min) each policy} \\ \lambda, s \\ \left\{ \begin{array}{l} \pi_{r2} \\ \tau_1, R_1 \\ \pi_{n1} \\ w_1 \\ \pi_{Grid1} \end{array} \right\} \\ p_{h1}, p_{l1} \end{array} \right.$	$\left\{ \begin{array}{l} \text{Max(Min) each policy} \\ \lambda, s \\ \left\{ \begin{array}{l} \pi_{r2} \\ \tau_2 \\ \pi_{n2} \\ w_2 \\ \pi_{Grid2} \end{array} \right\} \\ p_{h2}, p_{l2}, R_2 \end{array} \right.$	$\left\{ \begin{array}{l} \text{Max(Min) each policy} \\ \lambda, s \\ \left\{ \begin{array}{l} \pi_{r3} \\ \tau_3 \\ \pi_{n3} \\ w_3 \\ \pi_{Grid3} \end{array} \right\} \\ p_{h3}, p_{l3} \end{array} \right.$

3.2.2. Equilibrium solutions of scenario 2

In scenario 2, the distribution grid determines the electricity reliability and selling prices of electricity in low- and high-load periods in the first stage. In the second stage, by playing a Nash game between the producers, the wholesale electricity price for each producer is determined simultaneously. In the third stage, the government sets the equilibrium of its subsidy and RPS under each policy using backward induction. Equilibrium solutions of scenario 2 are given in Theorem 2.

Theorem 2. In scenario 2, the optimal decisions on the wholesale prices of conventional and renewable electricity, the reliability level, and the distribution grid's selling prices in low-load and high-load periods are given in Eqs. (22)–(26).

$$p_{l2} = \frac{(\mu^2 - 2\beta\psi)\alpha_l + H1(H2 + H3\theta_1 - H4\theta_2) - \alpha_h(\mu^2 + \psi(\theta_1 + \theta_2))}{H5(2\beta + \theta_1 + \theta_2)} \quad (22)$$

$$p_{h2} = -\frac{-(\mu^2 - 2\beta\psi)\alpha_h - H1(H2 - H7\theta_1 + H6\theta_2) + \alpha_l(\mu^2 + \psi(\theta_1 + \theta_2))}{H5(2\beta + \theta_1 + \theta_2)} \quad (23)$$

$$R_2 = -\frac{\mu(\alpha_h + \alpha_l + H1(2\beta - \theta_1 - \theta_2))}{H5} \quad (24)$$

$$w_2 = \frac{(M3 + (\alpha_l H8 - \alpha_h M2 + 2s\lambda H9)\theta_2 + H10\theta_2^2 + 2(-1 + \lambda)c_n M1 + \lambda c_r M1)}{3(\lambda - 1)M1} \quad (25)$$

Where the values of H_1 – H_9 and M_1 – M_5 are given in Appendix C.

3.2.3. Equilibrium solutions of scenario 3

In scenario 3, the distribution grid determines the selling prices of electricity in low- and high-load periods in the first stage. In the second stage, by playing a Nash game between the producers, the wholesale electricity price for each producer is determined simultaneously. In the third stage, the government sets the equilibrium of its subsidy and RPS under each policy using backward induction. Equilibrium solutions of scenario 3 are given in Theorem 3.

Theorem 3. In scenario 3, the optimal decisions on the wholesale prices of conventional and renewable electricity, the reliability level, and the distribution grid's selling prices in low-load and high-load periods are presented in Eqs. (27)–(30).

$$p_{l3} = -\frac{2\beta(-\alpha_l + E2(\beta - \theta_2)) - (\alpha_h + E2(-\beta + \theta_1))(\theta_1 + \theta_2)}{E1} \quad (27)$$

$$p_{h3} = -\frac{2E3\beta^2 + 2\beta\alpha_h + \theta_2(-E2\beta + \alpha_l + E2\theta_2) + \theta_1(\alpha_l + E2(\beta + \theta_2))}{(-2\beta + \theta_1 + \theta_2)(2\beta + \theta_1 + \theta_2)} \quad (28)$$

$$w_3 = -\frac{2s\beta^2\lambda - 2\beta^2\lambda c_r + E4 + E5\theta_2 - 4(-1 + \lambda)c_n(\beta^2 - \theta_1\theta_2)}{6(-1 + \lambda)(\beta^2 - \theta_1\theta_2)} \quad (29)$$

$$\tau_3 = \frac{2s\beta^2\lambda(1 - 3\rho) + 4\beta^2\lambda c_r + E4 + E6\theta_2 + 2(-1 + \lambda)c_n(\beta^2 - \theta_1\theta_2)}{6\lambda(\beta^2 - \theta_1\theta_2)} \quad (30)$$

Where the values of E_2 – E_6 are given in Appendix C.

Table 4

Parameter values.

par	unit	value	par	unit	value	par	unit	value
α_h	Mwh	8000000	Q_1	Mwh/ Ton CO_2	0.1	L_E	Ton CO_2	1.2×10^6
α_l	Mwh	7000000	Q_2	Mwh/ Ton CO_2	1.5	L_S	\$	2.5×10^{12}
β	Mwh/ \$	1	ρ	—	0.4	L_R	\$	3.2×10^{13}
θ_1	Mwh/ \$	0.3	c_r	Mwh/\$	6×10^5	f	—	0.6
θ_2	Mwh/ \$	0.2	c_n	Mwh/\$	5×10^5	φ	/Mwh \$	2×10^7
μ		0.4	ψ	—	1			

4. Numerical example and equilibrium solutions

This section presents a numerical example to demonstrate the feasibility of the mathematical model and examine the influence of essential parameters on the demands, pricing policies, and profits of conventional and renewable electricity producers and the distribution grid. Here, the parameters' values of the problem are set such that the

objective functions remain concave, and demands and prices remain positive. The values chosen for the parameters are provided in Table 4.

Figs. 3–5 (a, b, c) show the equilibrium solutions of the numerical example for each government policy. For all three policies, the feasibility region is determined using the intersection of the objective function with the two constraints. The functions related to social welfare, government revenue, and environmental impacts are drawn in purple, gray, and red, respectively. Each policy's objective function value is drawn as a contour plot. The points at the intersection of the constraint regions with the government objective function are the equilibrium points of the subsidy amount and the RPS ratio for each policy. In order to clarify Figs. 3–5, Fig. 3b is described as an example. In Fig. 3b, based on the diagram legend, the contour plot of the social welfare function has the lowest value at point A and the highest value at point B. According to the two constraints considered for the government revenue-generating functions and the adverse environmental effects, a feasible region has been obtained based on the sharing area between the contour plot and the constraints. The purple dashed line is related to the revenue-generating function of the government, and the red dashed line is related to the adverse environmental effects function. The intersection of the constraints with the contour plot is the equilibrium point for the

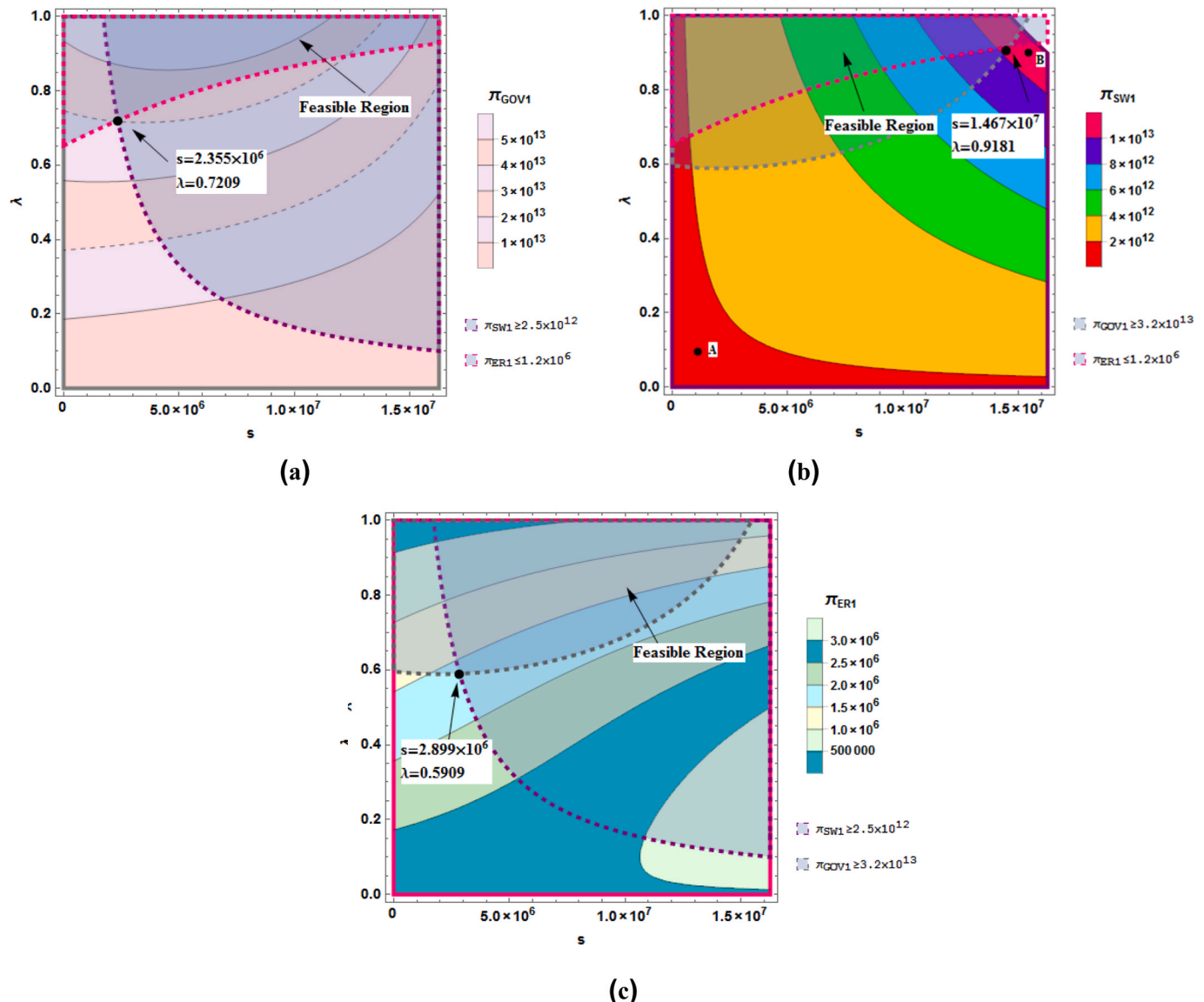


Fig. 3. Optimal equilibrium solution for government a) policy 1, (b) policy 2 and (c) policy 3 under scenario 1.

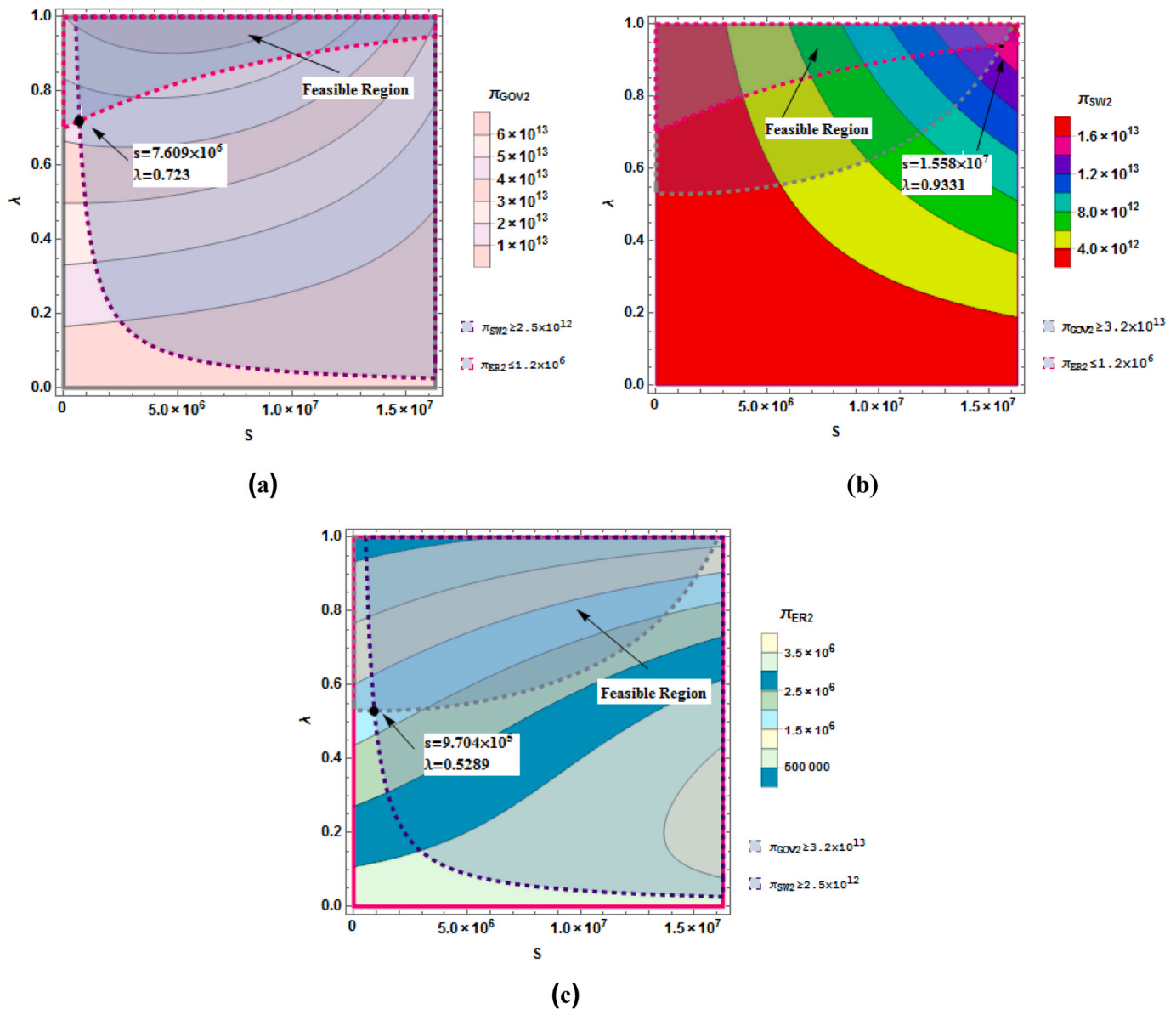


Fig. 4. Optimal equilibrium solution for government a) policy 1, b) policy 2 and c) policy 3 under scenario 2.

government's decision variables. At this point, the social welfare obtained by considering the lower limit for the revenue-generating function of the government and the upper limit for the adverse environmental effects and the feasible region has the most optimal value.

The optimal equilibrium solutions for each government policy in scenario 1 are presented in Tables 5 and 6. The solutions for scenario 2 are provided in Tables 7 and 8. Equilibrium values of decision variables in scenario 2, and the ones for scenario 3 are illustrated in Tables 9 and 10.

5. Discussion and results

Figs. (6–8) are drawn using the results tabulated in Table 5–10 for easier comprehension. These results can help government policymakers and electricity producers to make better decisions regarding their strategies.

Observation 1. As shown in Fig. 6(a), which displays the optimal RPS ratio for each policy in each scenario, the largest RPS belongs to policy 2 in scenario 2.

Observation 2. Fig. 6 (b) shows each scenario's optimal subsidy amount for each government policy. Based on the subsidy amount, the highest value belongs to policy 2 in scenario 2, and the lowest belongs to policy 1 in the same scenario.

Insight 1. In scenarios where the distribution grid invests in energy storage technologies, the government should pursue the maximum RPS and subsidization strategy if its goal is maximizing social welfare. In contrast, the government should adopt the minimum subsidy amount if its goal is revenue maximization.

Observation 3. Fig. 7 (a, b, c) show the equilibrium profit of the renewable electricity producer, the conventional electricity producer, and the distribution grid for each policy in each scenario. Both electricity producers and the distribution grid achieve the most profit in policy 2 and scenario 2.

Insight 2. When the goal is the maximization of the profits of supply chain members, the government should pursue the policy of maximizing social welfare, and the distribution grid should invest in energy storage technologies for reliability improvement to profit.

Observation 4. Fig. 8 shows the optimal reliability for each policy in each scenario. When $\mu = 0.4$, the highest reliability belongs to policy 2 in

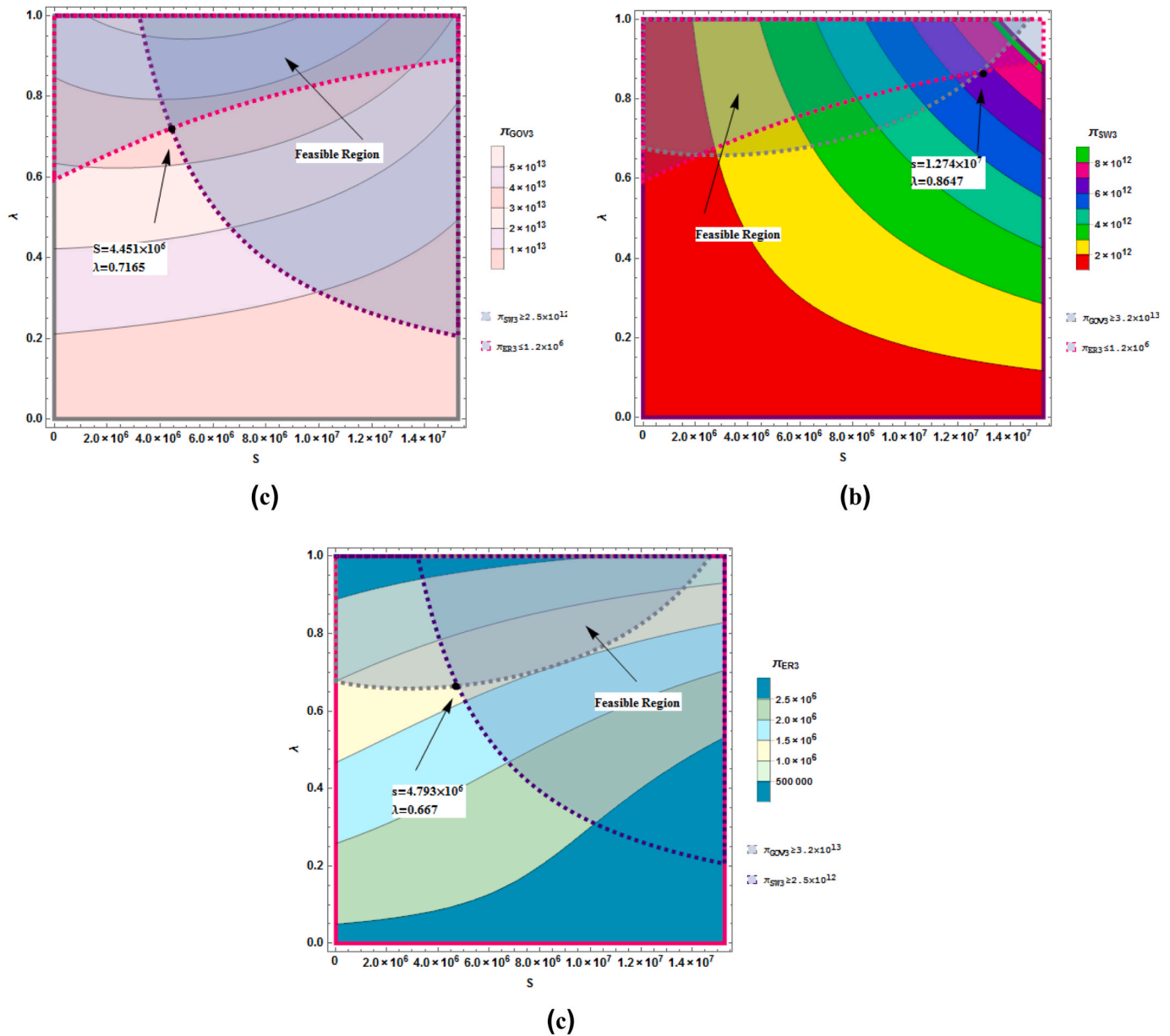


Fig. 5. Optimal equilibrium solution for government a) policy 1, b) policy 2 and c) policy 3 under scenario 3.

Table 5

Equilibrium values of decision variables in scenario 1.

	Policy 1	Policy 2	Policy 3
s_1	2.36E+06	1.47E+07	2.90E+06
λ_1	0.7209	0.9181	0.5909
R_1	2.80E+06	4.67E+06	2.81E+06
τ_1	5.02E+06	1.55E+06	6.00E+06
w_1	1.56E+07	2.00E+07	1.08E+07
p_{h1}	9.45E+06	8.86E+06	9.44E+06
p_{l1}	9.34E+06	8.40E+06	9.33E+06
D_{h1}	1.54E+06	3.18E+06	1.55E+06
D_{l1}	1.62E+06	3.33E+06	1.62E+06

scenario 1. In other words, when producers jointly invest in energy storage technologies (scenario 1), the reliability of the electricity supply under all three government policies is higher than when the distribution grid makes investments (scenario 2). The effect of the reliability sensitivity coefficient on reliability is displayed in Fig. 9. This figure shows that as the sensitivity coefficient increases, higher reliability is created in scenario 2 than in scenario 1.

Table 6

Equilibrium values of functions in scenario 1.

	Policy 1	Policy 2	Policy 3
π_{r1}	1.09E+13	4.95E+13	1.10E+13
π_{n1}	1.17E+13	2.58E+13	1.18E+13
π_{Grid1}	6.65E+12	1.76E+13	6.69E+12
π_{Gov1}	4.02E+13	3.18E+13	3.20E+13
π_{sw1}	2.50E+12	7.31E+12	2.52E+12
π_{ER1}	1.20E+06	1.183E+06	1.181E+06

Observation 5. Based on Fig. 10 (a, b), in the scenario that does not invest in storage technologies (Scenario 3), the prices are lower in periods of low and high load than in the other two scenarios. Therefore, it is expected that the demand under the influence of prices in scenario 3 is more than in the other two scenarios. However, due to the influence of demand from the level of reliability, the demand in both periods of low and high load in the other two scenarios has exceeded scenario 3 (Fig. 10 (c,d)). Comparison of Fig. 10(a) with (b) and Fig. 10(c) with (d) shows

Table 7

Equilibrium values of decision variables in scenario 2.

	Policy 1	Policy 2
s_2	7.610E+05	1.560E+07
λ_2	0.723	0.9331
R_2	1.690E+06	3.308E+06
τ_2	5.343E+06	1.391E+06
w_2	9.537E+06	1.806E+07
p_{h2}	8.855E+06	7.759E+06
p_{l2}	8.722E+06	7.410E+06
D_{h2}	1.565E+06	3.064E+06
D_{l2}	1.610E+06	3.152E+06

Table 8

Equilibrium values of functions in scenario 2.

	Policy 1	Policy 2	Policy 3
π_{r2}	1.19E+13	5.88E+13	8.87E+12
π_{n2}	1.05E+13	3.97E+13	1.07E+13
π_{Grid2}	5.28E+12	2.02E+13	5.30E+12
π_{Gov2}	3.89E+13	3.20E+13	3.84E+13
π_{SW2}	2.52E+12	8.00E+12	2.53E+12
π_{ER2}	1.50E+06	1.188E+06	1.174E+06

Table 9

Equilibrium values of decision variables in scenario 3.

	Policy 1	Policy 2
s_3	4.45E+06	1.27E+07
λ_3	0.7165	0.8647
R_3	0.00E+00	0.00E+00
τ_3	3.80E+06	8.26E+05
w_3	1.53E+07	5.09E+07
p_{h3}	7.98E+06	6.73E+06
p_{l3}	7.81E+06	6.46E+06
D_{h3}	1.58E+06	2.56E+06
D_{l3}	1.58E+06	2.56E+06

Table 10

Equilibrium values of functions in scenario 3.

	Policy 1	Policy 2	Policy 3
π_{r3}	1.33E+13	3.49E+13	1.33E+13
π_{n3}	1.33E+13	3.49E+13	1.33E+13
π_{Grid3}	6.65E+12	1.71E+13	6.67E+12
π_{Gov3}	3.52E+13	3.209E+13	3.21E+13
π_{SW3}	2.50E+12	6.57E+12	2.51E+12
π_{ER3}	1.21E+06	1.20E+06	1.162E+06

that the use of the Time-Of-Use pricing method leads to the reduction of consumption in peak periods with peak prices, and as a result, it seeks demand-side management. Also, based on these figures, although the demand from the distribution grid in scenario 1 is more than in scenario 2; however, the profit of the distribution grid in scenario 2 is more than in scenario 1 (Fig. 7(c)).

The effects of government decision variables on demand, wholesale renewable and conventional electricity prices, government revenue, adverse environmental impacts, and social welfare are illustrated in Figs. 11–15. Wherever we performed sensitivity analysis on parameters or other variables, we considered other variables in the optimal state based on the relevant scenario. As these figures show, government decisions have a major effect on prices, demand, environmental impacts, and social welfare. The analyzes and results presented in this study can help governments, regardless of the region and country, to make the right decisions in the policies of the energy sector.

Observation 6. As shown in Fig. 11 (a) and (b), the increasing in the RPS and subsidy amount increase the demand of the distribution grid for renewable power while its demand for non-renewable (conventional) power decreases. The increase is more pronounced in the scenario where the distribution grid invests in energy storage technologies (Scenario 2).

Insight 3. Based on the numerical example provided, given the impact of RPS and subsidies on customers' motivation to purchase electricity from renewable producers, to reach the goals of sustainable development, the government should strive to move toward the RPS ratio of 1 while paying more subsidies to increase demand in the distribution grid for renewable power. But considering that it is difficult to achieve 100% of RPS in areas where the application of energy storage technology has not matured, based on the limitations of each country, the government of that country should adopt an appropriate value for RPS.

Observation 7. As shown in Figs. 12 and 13 (a, b), wholesale prices in scenario 2 are lower than scenario 1. The wholesale prices of the conventional producers increase by increasing RPS and subsidies, leading to a decrease in demand for non-renewable power (Fig. 11). On the contrary, the wholesale prices of the renewable electricity producers decrease by increasing RPS and subsidies, resulting in higher purchases from this producer (Figs. 12 and 13 (a)).

Insight 4. Government decisions significantly affect the prices of conventional and renewable electricity producers and, consequently, the demand for renewable and non-renewable electricity which can be employed to pursue sustainable development goals in this area.

Fig. 14 (a, b, c) and 15 (a, b) show the changes in the value of objective functions with changes in RPS and subsidy amount in three proposed scenarios.

**Fig. 6.** Equilibrium values of (a) RPS and (b) subsidy amount for the three government policies in the three scenarios.

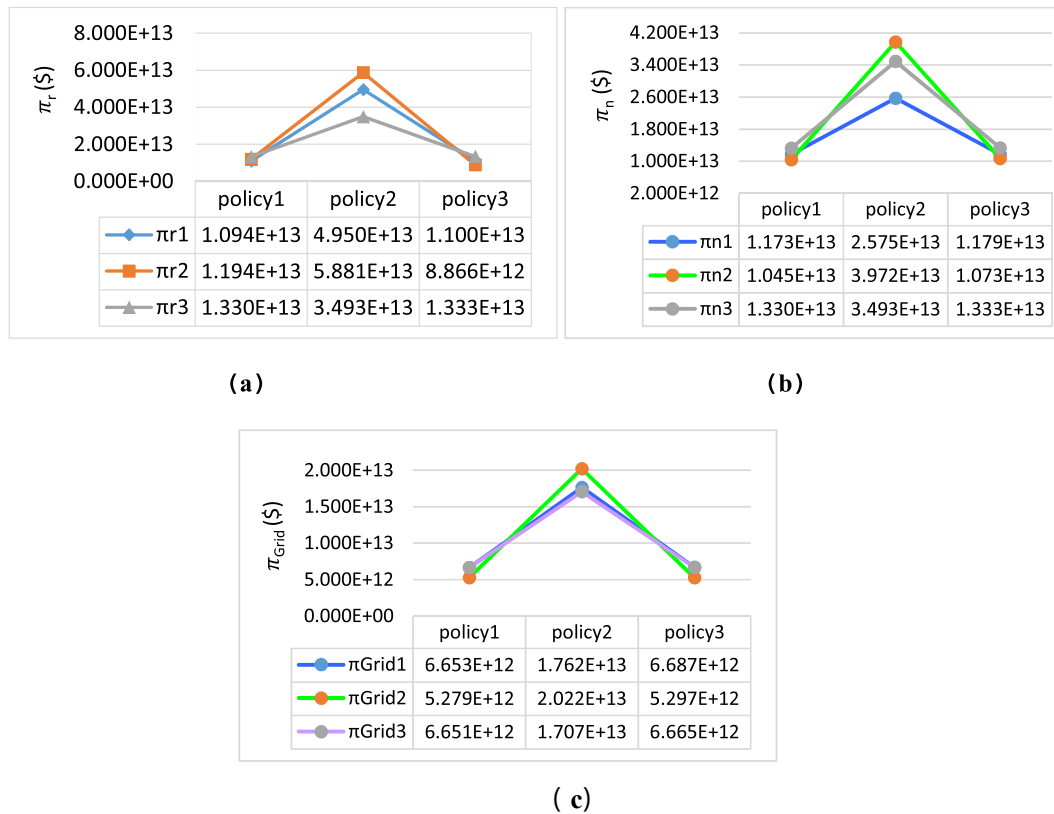


Fig. 7. Equilibrium values of the profit functions of (a) the renewable electricity producer (b) the conventional electricity producer, and (c) the distribution grid.

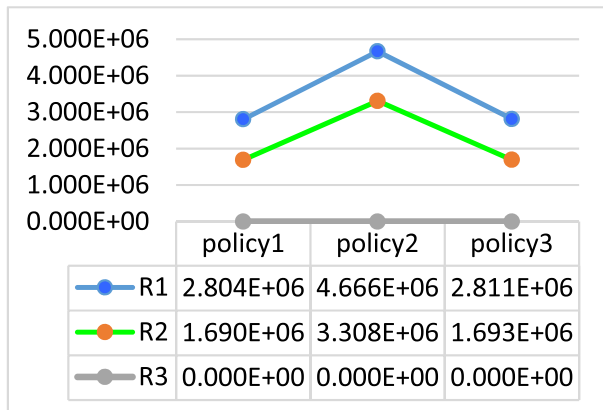


Fig. 8. Equilibrium reliability.

Observation 8. As Fig. 14 (a, b, c), an increase in RPS leads to an increase in government revenue and social welfare while a significant decrease in the adverse environmental impacts. As shown in Fig. 15 (a), as the government subsidy increases, the government revenue has an increased and then a decreasing trend. Social welfare increases with increasing subsidies (Fig. 15(b)). Overall, by variation in RPS and subsidy amount, scenario 2 improves government functions with a greater slope compared to scenarios 1 and 3.

Insights 5. To achieve sustainable development goals like maximizing social welfare and minimizing adverse environmental impacts, the government must commit to paying high amounts of subsidies and enforcing higher RPS ratios. Also, considering the best performance of scenario 2 among its counterparts, it is recommended to work on making the distribution grid, the main investor in energy storage technologies, for reliability improvement.

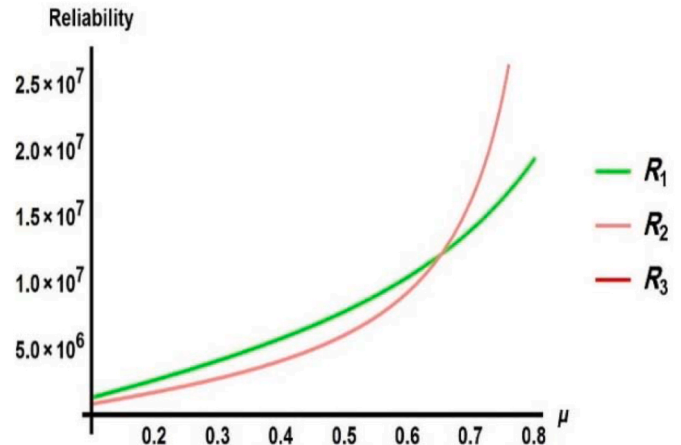


Fig. 9. Reliability changes with changes in its sensitivity coefficient.

Observation 9. Fig. 16 (a) shows the changes in the reliability sensitivity coefficient for the consumers' demand from the distribution grid in low- and high-load periods. In scenario 2, the reliability sensitivity coefficient changes influence the demands for low- and high-load periods. By increasing the sensitivity coefficient, the demand for the low-load period increases more than the high-load period. Fig. 16 (b) shows the effects of the reliability and load shift sensitivity coefficients on consumer demand from the distribution grid. As illustrated in this Fig., by increasing these coefficients, demand shifts from the high-load period to the low-load period.

Insight 6. According to observation 9, in the scenario where the distribution grid invests in energy storage technologies for reliability improvement, demand increases more with increased reliability compared to other scenarios. Notably, as reliability increases, the

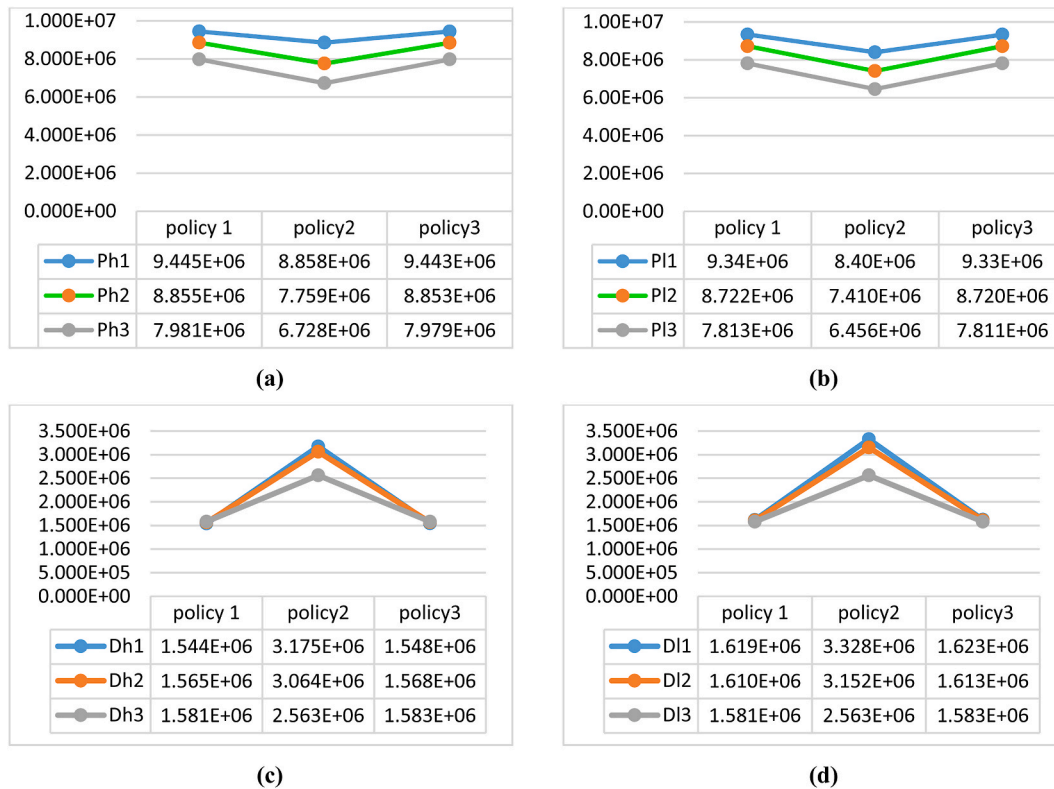


Fig. 10. Equilibrium values a) price in high-load period b) price in the low-load period c) demand in the high-load period d) demand in the low-load period.

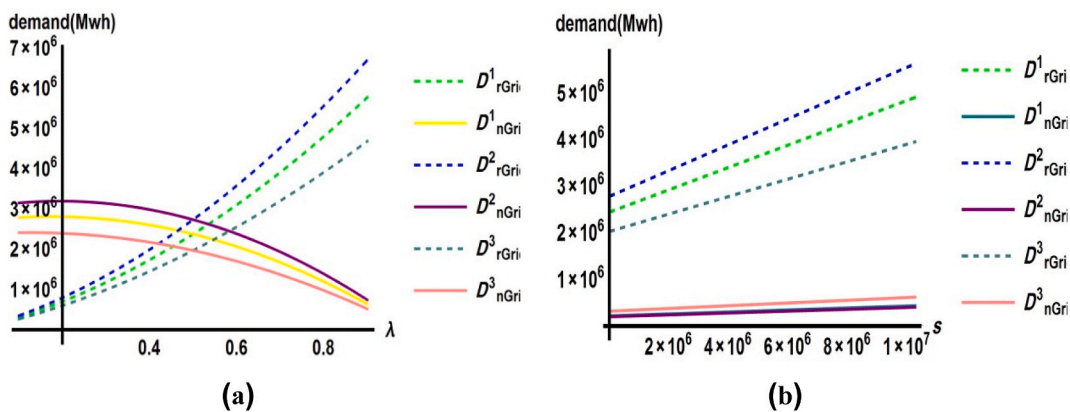


Fig. 11. Changes in the distribution grid demand from producers with changes in a) RPS ratio and b) subsidy.

demand in the off-peak period increases more than the high-peak period. Also, in this scenario, adjustment of the parameters leads to demand-side management.

6. Conclusion

This paper examined the Time-Of-Use pricing in an electricity supply chain with government intervention for reliability improvement using a game-theoretic approach. Supply chain members were assumed to include a renewable electricity producer, a conventional electricity producer, the distribution grid, and consumers. Demand functions representing the amount of electricity purchased from the distribution grid were developed by incorporating reliability levels and demand-side management in low- and high-load periods. In a leader-follower game structure, the government was considered the leader of the game, electricity producers in competition with each other were considered the

first follower, and the distribution grid was considered the second follower. It was assumed that the government follows three policies to achieve its sustainable development objectives, including maximizing its revenue and social welfare and minimizing adverse environmental effects. Three investment scenarios were defined for the supply chain model, and equilibrium solutions of each scenario for each of the three government policies were obtained. The impacts of government decision variables, including subsidies and Renewable Portfolio Standards, on the wholesale prices of renewable and conventional electricity producers, distribution grid's demand from producers, end-consumer demand and prices, and government functions in all three scenarios were examined.

According to the numerical example and sensitivity analysis results, when the government follows the social welfare maximization policy, the prices in both periods of low- and high-load will be lower than other policies. In the scenario where the distribution grid invests in storage technologies, the prices will be lower than in scenario 1. In scenario 3,

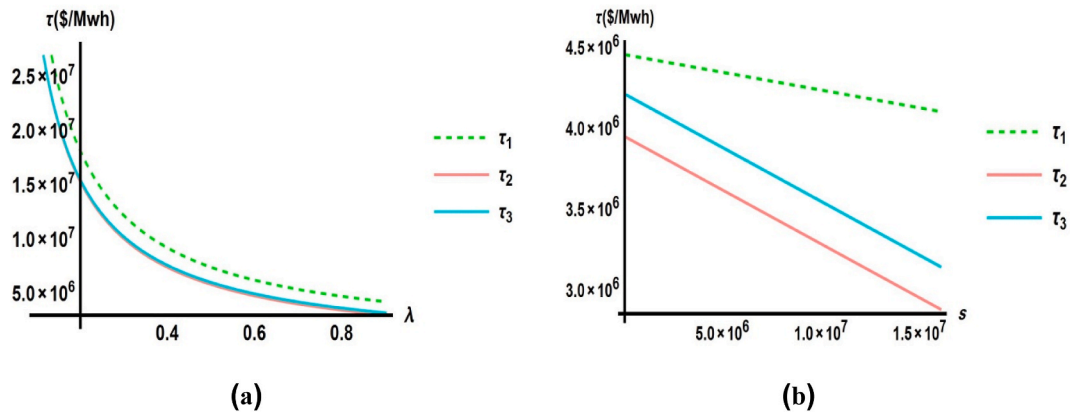


Fig. 12. Changes in the wholesale price of renewable electricity with changes in (a) RPS ratio and (b) subsidy.

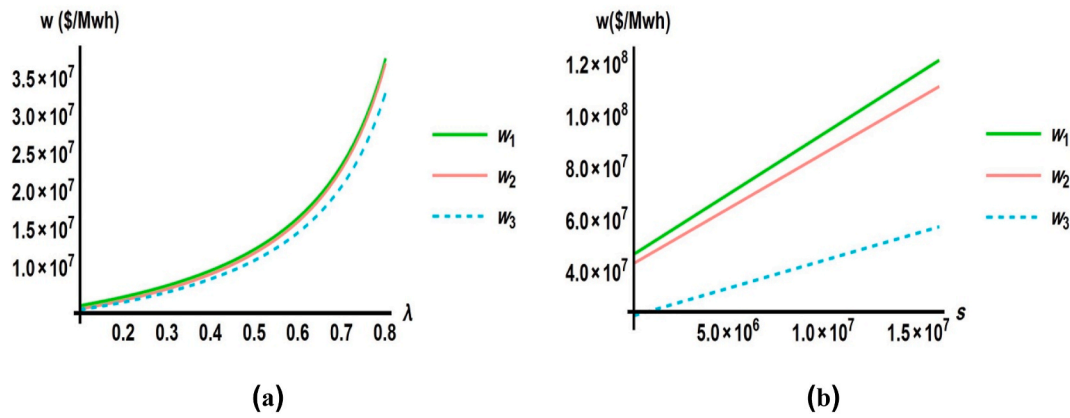


Fig. 13. Changes in the wholesale price of non-renewable (conventional) electricity with changes in a) subsidy and b) RPS ratio.

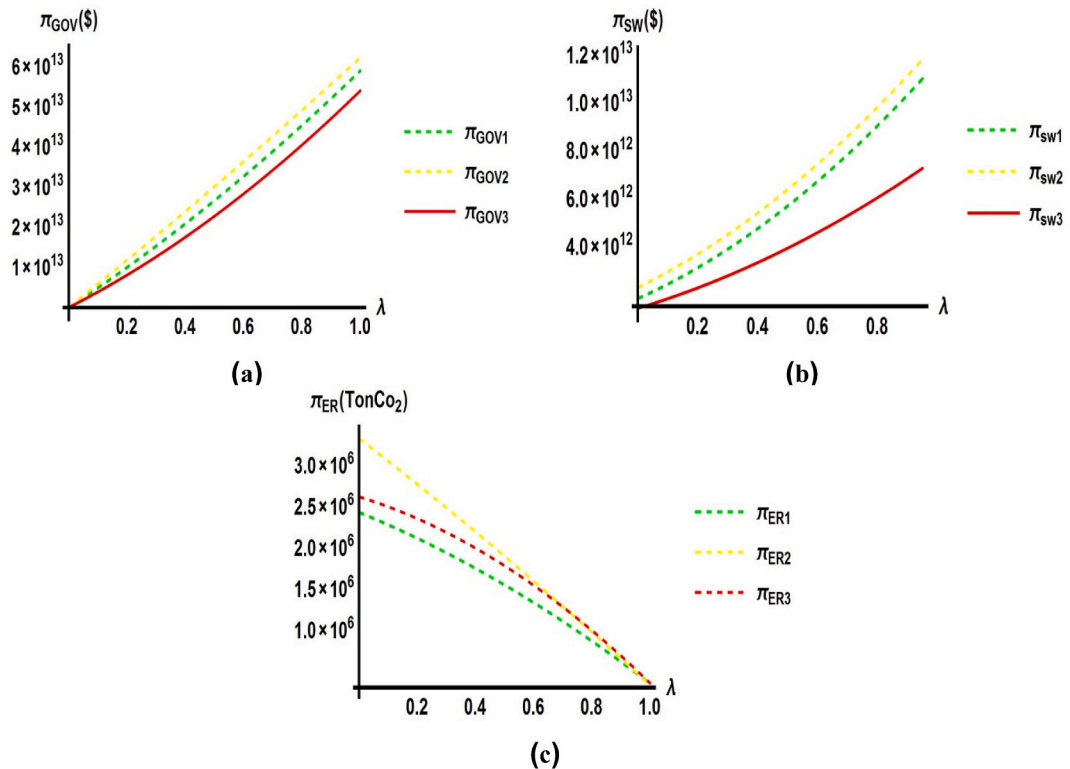


Fig. 14. Changes in (a) the government revenue function, (b) the social welfare function, and (c) the environmental impacts function with changes in RPS ratio.

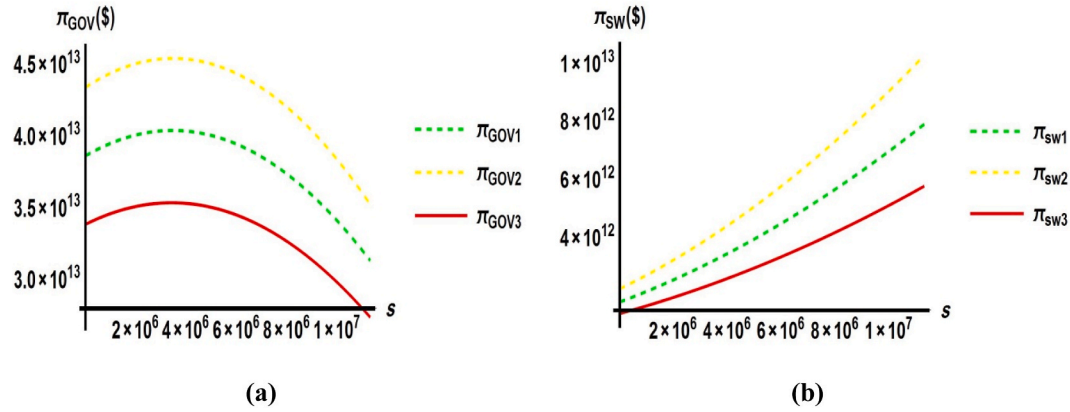


Fig. 15. Changes in (a) the government revenue function, (b) the social welfare function with changes in subsidy.

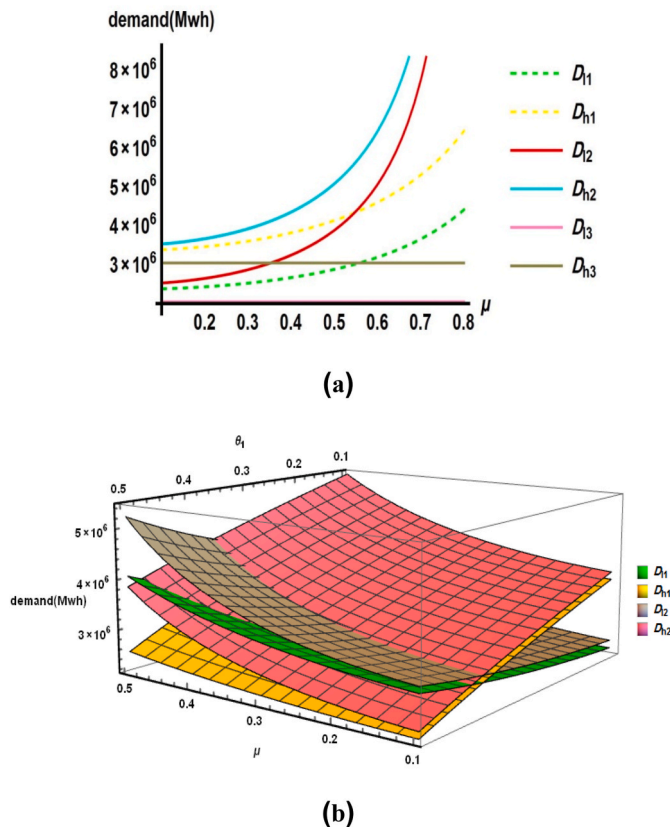


Fig. 16. Changes in consumer demand versus changes in a) reliability sensitivity coefficient and b) θ_1 and reliability sensitivity coefficient.

because the cost for investment is not done, the prices are lower than in other scenarios. Under the influence of these prices, the demand in both periods of low and high load under the social welfare maximization policy has increased more than the other two policies. Also, due to the influence of the demand from the reliability level, the demand in the other two scenarios has increased more than in scenario 3. In addition to these, using the Time-Of-Use pricing method leads to the reduction of consumption in peak periods with peak prices, and as a result, it seeks demand-side management. Also, the selection of scenario 2 by the supply chain members, in addition to further increasing the reliability and profit for producers and the distribution grid, helps governments achieve the goals of maximizing social welfare and minimizing adverse environmental effects more than the other two scenarios.

Nevertheless, if the goal is to maximize the government's revenue

generation, choosing scenario 1 is a better strategy for the government. Also, the government should pay lower subsidies if the goal is revenue maximization. In contrast, the government must pay higher subsidies and enforce higher Renewable Portfolio Standards ratios if it aims to support the production and renewable energy applications to maximize social welfare and minimize adverse environmental impacts. Furthermore, the profits of supply chain members and the electricity supply reliability were higher in the social welfare maximization policy than in the other two policies. Based on this policy, high prices of conventional and low prices of renewable energy lead to exceeding the demand for renewable electricity over the demand for conventional electricity.

Considering the findings of this study regarding the significant role of government goals and strategies in the decisions of supply chain members, including prices, profits, and supply reliability levels, these findings can help governments and electricity producers, regardless of the region and country, to choose better strategies in accordance with their goals when interacting in an electricity supply chain with renewable sources.

Based on the presented results, the findings that help the government and other decision-makers of the electricity supply chain are presented as follows.

- Due to the significant impact of government strategies such as subsidies and Renewable Portfolio Standards on wholesale prices, demand from producers, and achieving sustainable development goals, the government of each country should consider the limitations of technology and the components of sustainable development of that country for Renewable Portfolio Standard should adopt a suitable amount.
- Although adopting a high amount for the subsidy has a positive effect on the social and environmental components, its excessive increase has a negative effect on the economic component of the government, so a moderate amount should be adopted for the subsidy.
- Following the social welfare maximization policy of the government compared to the other two policies creates more utility for the producers' profit, the distribution grid, and the reliability improvement.
- It is more desirable to follow the policy of maximizing social welfare by the government in order to achieve the goal of reducing prices in low and high periods. Also, using the Time-Of-Use pricing method makes consumption in peak periods with peak prices lower than consumption in off-peak periods.
- The selection of scenario 2 by the supply chain members helps governments achieve the goals of maximizing social welfare and minimizing adverse environmental effects more than the other two scenarios. However, if the goal is to maximize the government's revenue generation, choosing scenario 1 is a better strategy for the government.

- Investment by the distribution grid is more favorable than the investment by producers to improve reliability and increase the profit of renewable and conventional producers and the profit of the distribution grid.

The following limitations and challenges in this research can be extended for future research: First, this research can be expanded by adding other government objectives, such as job creation and energy security, to develop renewable sources. Second, in this research, it is assumed that subsidies are given to the distribution grid and the renewable producer to promote the production and consumption of renewable energy. Future research suggests that the subsidy is only given to one supply chain member (for example, the renewable energy producer or the distribution grid) and compares the results under different scenarios. Third, considering the growing number of electricity-producing companies, it is recommended to examine the long-term effect of policies on the decisions of larger groups of these companies and how they respond to different strategies using the evolutionary game-theoretic approach. Fourth, in this research, the cooperation between producers is considered to improve the reliability of electricity supply, considering the cooperation between producers and the distribution grid in the form of contracts is also an attractive topic for future research. Finally, incorporating other factors, such as service quality and public awareness, into demand functions for renewable power can also be a good path for future research.

Credit author statement

Sima Amiri-Pebdani: Conceptualization, Methodology, Software, Data curation, Writing e original draft, Visualization, Investigation. Mahdi Alinaghian: Supervision, Validation, Writing - review & editing. Hossien Khosroshahi: Validation, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2023.128721>.

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